



PMA-213

TACAN Shore Based Antenna Group Performance Specification

15 January 2026

Reviewers/Approvals

Signatory

Date

Submitted:	_____	_____
	TACAN Lead Engineer	
Approved:	_____	_____
	Ship Air Traffic Management IPT Lead, PMA-213	
Approved:	_____	_____
	Product Support Manager, PMA-213	
Approved:	_____	_____
	Assistant Program Manager T&E, PMA-213	
Approved:	_____	_____
	Assistant Program Manager Engineering, PMA-213	

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1. INTRODUCTION

1.1. SCOPE

This specification establishes the performance, design, test, manufacture and acceptance requirements for the TACAN Antenna System. This specification also describes a TACAN Antenna System consisting of the following two major functional units: an electronically scanning antenna and a signal generating function.

1.2. OVERVIEW

The antenna system described herein utilizes a vertically polarized antenna capable of transmitting frequency (RF) energy in the 960 MHz to 1215 MHz frequency band.

2. APPLICABLE DOCUMENTS

2.1. GENERAL

The documents listed in this section are specified in this specification. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in this specification, whether or not they are listed.

2.2. GOVERNMENT DOCUMENTS

The following Government documents, drawings, and publications of the exact revision level shown form a part of this document to the extent specified herein.

MIL-HDBK-1857	Grounding, Bonding and Shielding Design Practices
MIL-STD-464D	Electromagnetic Environmental Effects, Requirements for Systems
MIL-HDBK-454B	General Guidelines for Electronic Equipment
MIL-STD-810H	Test Method Standard, Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1399/300B	Electric Power, Alternating Current
MIL-STD-461G	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-130N	Identification Marking of U.S. Military Property
MIL-STD-291C	Standard Tactical Air Navigation (TACAN) Signal
MIL-STD-1472G	Human Engineering

2.3. NON-GOVERNMENT PUBLICATIONS

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Institute of Electrical and Electronics Engineers (IEEE)

C95.1 Standard for Safety Levels with Respect to Human Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz (2019)

2.4. ORDER OF PRECEDENCE

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. SYSTEM CHARACTERISTICS

- a. The antenna system shall meet the TACAN signal requirements in MIL-STD-291C
- b. The antenna system shall interface with the existing AN/URN-32 TACAN Transponder without requiring changes to the existing cabling as shown in Attachment 1, AN/URN-32 Cable Running Sheets.
- c. The antenna system in conjunction with the existing TACAN transponder (AN/URN-32) shall receive interrogations from aircraft and transmit replies that provide azimuth and distance information to aircraft
- d. The antenna system shall utilize integral monitoring to provide signals to the TACAN transponder when a fault in the antenna system exists.
- e. The antenna system shall operate without modifications within the frequency band from 960 to 1215 MHz.
- f. The antenna system shall modulate the 15 Hz and 135 Hz components of the TACAN transponder pulse train, which in conjunction with the reference burst pulse groups provides the coarse (15 Hz) and fine (135 Hz) azimuth information signals.
- g. North and auxiliary reference trigger pulses shall be generated by the TACAN antenna system and sent to the TACAN transponder equipment for the purpose of generating reference pulses.
- h. The TACAN antenna system shall establish a vertical radiation pattern that provides the coverage requirements described in 3.2.2.4.
- i. The antenna system shall be capable of operating in either TACAN mode or Distance Measuring Equipment (DME)-only mode.

3.2. PERFORMANCE REQUIREMENTS

3.2.1. ELECTRICAL INTERFACE WITH AN/URN-32 TACAN TRANSPONDER

The antenna system shall operate with the AN/URN-32 TACAN Transponder using existing interconnecting cables for shore stations as shown in Attachment 1.

3.2.1.1. NORTH TRIGGER

The North trigger shall interface with the TACAN Transponder as shown in Attachment 1.

3.2.1.2. AUXILIARY TRIGGER

The Auxiliary trigger signal shall interface with the TACAN transponder as shown in Attachment 1.

3.2.1.3. ANTENNA SHUTDOWN ALARM SIGNAL

- a. The antenna shutdown alarm signal shall interface with the AN/URN-32 TACAN transponder as shown in Attachment 1. This signal indicates the alarm status of the antenna system.
- b. The shutdown alarm shall be provided to the TACAN transponder when a fault that affects azimuth accuracy has been detected by the system monitoring equipment.

3.2.1.4. ANTENNA SOFT ALARM

- a. The maintenance alert signal shall interface with the AN/URN-32 TACAN transponder as shown in Attachment 1.
- b. The soft alarm shall provide an indication and a soft alarm signal to the TACAN transponder of any faults detected in the antenna system that require maintenance.

3.2.1.5. TACAN/DME-ONLY MODE

- a. The TACAN/DME-only signal shall interface with the TACAN transponder as shown in Attachment 1.
- b. When the antenna system is in DME-only mode, the TACAN modulation (3.2.4) shall be removed.

3.2.1.6. EQUIPMENT GROUND

This antenna system ground shall interface with the TACAN transponder as shown in Attachment 1.

3.2.2. ANTENNA PATTERN

3.2.2.1. POLARIZATION

- a. The antenna system shall be capable of radiating and receiving vertically polarized signals.
- b. Any horizontally polarized signal shall be at least 26 dB less than the vertically polarized signal.

3.2.2.2. ANTENNA GAIN

With the radiated signal being modulated, the resulting carrier pattern shall have a peak gain of at least 8.0 dBi, where $\text{Gain(dBi)} = D - \text{IL}$, Gain in dBi is the antenna's directivity minus its insertion loss.

3.2.2.3. HORIZONTAL FIELD PATTERN

- a. The antenna shall have an omnidirectional radiation pattern.
- b. When measured on the horizon, with the TACAN modulation on, the time averaged power density measured over all azimuth angles shall not vary by more than ± 1 dB.

3.2.2.4. VERTICAL FIELD PATTERN

With TACAN modulation removed, the vertical field pattern shall be in accordance with 3.2.2.4.1 through 3.2.2.4.5 at all azimuths.

3.2.2.4.1. MAIN LOBE

- a. The main lobe of energy in the vertical plane shall be 5.0 degrees wide or more at the half power points.
- b. The vertical angle of maximum radiated energy shall be between 2.0 degrees and 7.0 degrees above the horizon.

3.2.2.4.2. SLOPE

The rate of change of the signal level from 1.0 degree below the horizon to 1.0 degree above the horizon shall be ≥ 0.20 V/V/degree when the vertical pattern data is normalized to 1.0 V at the horizon.

3.2.2.4.3. GAIN ON THE HORIZON

The gain on the horizon shall be at least 2 dBi.

3.2.2.4.4. GAIN BELOW THE HORIZON

The gain at angles between 6 and 50 degrees below the horizon shall be less than the gain at the peak of the major lobe above the horizon by at least 16 dB. [$\text{Gain}_{6-50\text{below}} \leq (\text{G}_{\text{peak}} - 16 \text{ dB})$]

3.2.2.4.5. GAIN ABOVE THE HORIZON

- a. The gain at angles between 6 and 20 degrees above the horizon shall be greater than the peak of the major lobe above the horizon minus 15 dB. [$\text{Gain}_{\text{horizon}} > (\text{Gain}_{\text{peak}} - 15 \text{ dB})$]
- b. The gain at angles between 20 and 50 degrees above the horizon shall be greater than the peak of the major lobe above the horizon minus 25 dB. [$\text{Gain}_{\text{horizon}} > (\text{Gain}_{\text{peak}} - 25 \text{ dB})$]

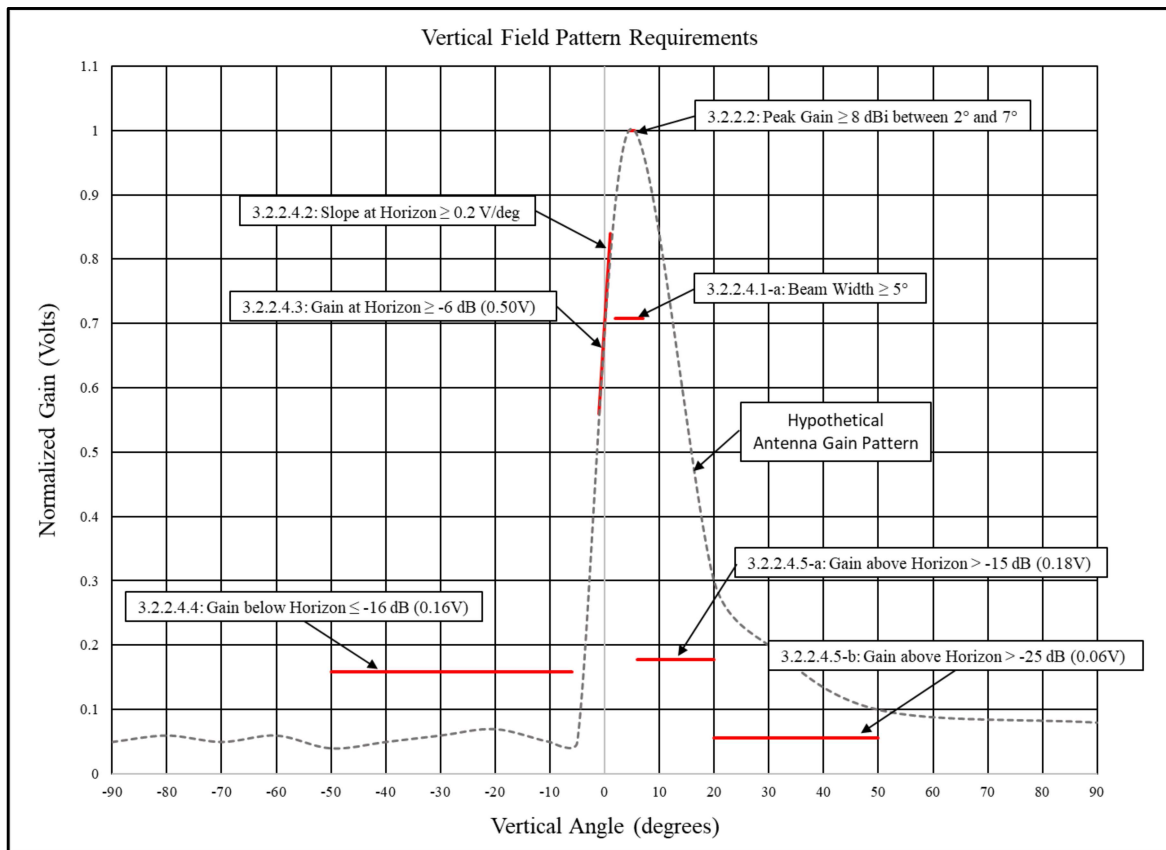


FIGURE 3-3-1 Vertical Gain Pattern Requirements with Hypothetical TACAN Antenna

3.2.3. AZIMUTH VARIABLE SIGNAL

- The field pattern generated by the antenna system shall rotate at a rate of 15 Hz $\pm$ 0.2%.
- This rotation shall produce a variable signal, nominally composed of 15 Hz and 135 Hz sine wave components.
- The 15 Hz and 135 Hz signals shall be synchronous with the rotation rate.

3.2.4. MODULATION

- The antenna system shall, in a region from 3 degrees below the horizon¹ to 20 degrees above the horizon, amplitude modulate the RF carrier, in the absence of harmonics, according to the following formula:

$$Y = 1.0 A \sin(2\pi f t \pm \theta - y) + B \sin(18\pi f t \pm 9\phi - 9y)$$

Where:

Y = normalized composite 15-135 Hz signal amplitude

A = percent of modulation of the 15 Hz component

B = percent of modulation of the 135 Hz component

θ = deviation in azimuth degrees from the coincidence of the 15 Hz modulation signal and the main reference pulse

ϕ = deviation in azimuth degrees from the coincidence of the 135 Hz modulation signal and the auxiliary or 40 degrees pulse

¹ Unless the antenna design mitigates the effects of longitudinal multipath to the extent necessary to achieve the required performance above the horizon (0° - 20°) in the presence of worst-case ground reflection multipath.

y = bearing to the antenna from the point of observation

f = pattern rotation in Hz

t = time in seconds from the corrected position of the 10th pulse of the X-mode main reference group or 15 μ sec after the 6th pulse of the Y-mode main reference group. t is computed from the average position of the 12th pulse of each auxiliary reference group.

With the following tolerances:

$f = 15 \text{ Hz} \pm 0.2\%$

$\theta = 3.5$ azimuth degree RMS with peak value not to exceed ± 8 azimuth degrees

$\phi = 0.5$ azimuth degree RMS with peak value not to exceed ± 2 azimuth degrees

$A = 21\% \pm 9\%$, except as noted in b below

$B = 21\% \pm 9\%$, except as noted in b below

- b. In the region from 3 degrees below the horizon to 3 degrees above the horizon, the mean percent of modulation of the 15 Hz and 135 Hz components shall each be $21\% \pm 4\%$.
- c. The cross polarization effects up to plus or minus 45 degrees from the vertical and within elevation angles from 6 degrees below the horizon to 40 degrees above the horizon, shall be limited such that θ does not deviate more than 3 degrees and ϕ does not deviate more than 1 degree.

3.2.5. REFERENCE BEARING SIGNALS

- a. There shall be two separate synchronized output of reference trigger pulses, the north trigger pulse and the auxiliary trigger pulse.
- b. The north triggers and auxiliary triggers shall be synchronized with the antenna pattern rotation.
- c. For each 360 degrees of rotation, one north trigger pulse shall be generated.
- d. Forty degrees after the north trigger pulse, an auxiliary pulse shall be generated at each of eight consecutive angular increments of $40 \text{ degrees} \pm 0.01 \text{ degree}$.
- e. The ninth 40 degree increment auxiliary trigger pulse, which would otherwise coincide in time with the north trigger pulse, shall not be generated.

3.2.5.1. TRIGGER PULSE SHAPE

Each reference trigger pulse shall occur as a positive half cycle followed by a negative half cycle.

3.2.5.2. TRIGGER PULSE RISE TIME

Pulse rise time shall be less than or equal to 30.0 microseconds

3.2.5.3. TRIGGER PULSE FALL TIME

Pulse fall time shall be less than or equal to 2.0 microseconds.

3.2.5.4. TRIGGER PULSE WIDTH

Each half cycle shall be between 90 microseconds and 100 microseconds in width, as measured between the 50% pulse amplitude points, as shown in Figure 3-2.

3.2.5.5. TRIGGER PULSE AMPLITUDE

- a. The main trigger pulse amplitude shall be +7.5 VDC to -7.5 VDC
- b. The maximum amplitude of extraneous voltages shall not exceed 0.20 V peak-to-peak as measured at the output terminals and connected to a load.

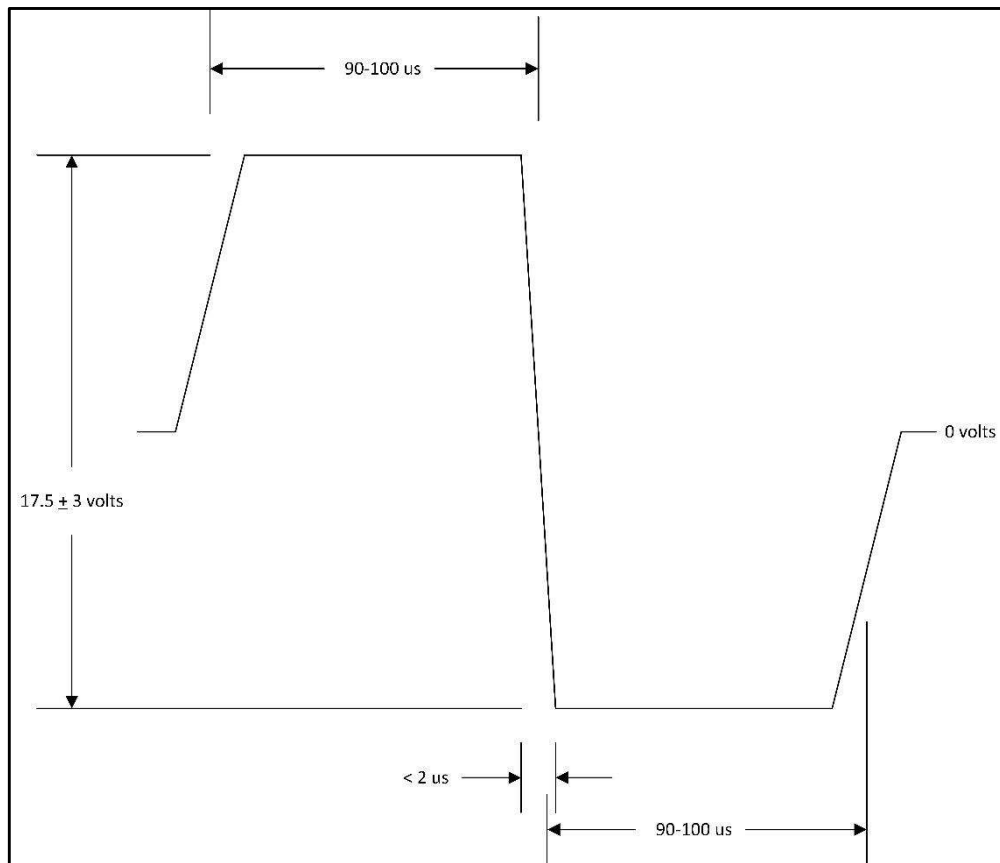


Figure 3-2 Trigger Pulse Shape

3.2.6. ANTENNA AZIMUTH ALIGNMENT

- a. The alignment shall be adjustable for at least ± 30 degrees in increments of 0.1 ± 0.05 degrees from the antenna control equipment located in the TACAN ground transponder equipment shelter.
- b. The alignment shall be accomplished by electronically changing the position of the 15 Hz and 135 Hz modulation envelope position relative to the position of the north and auxiliary reference triggers.
- c. The rotation control used to accomplish the alignment shall be capable of being set to indicate 00.0 when the magnetic north information being radiated by the antenna is oriented physically to magnetic north.
- d. A visual reading of the control setting shall be provided with the same resolution as the setting.
- e. This rotation for magnetic north alignment shall not change the position of the north reference triggers relative to the auxiliary reference triggers.
- f. Additionally, the phase relationships of the 15 Hz modulation envelope, relative to the 135 Hz modulation shall not be changed during this adjustment.

3.2.7. RF POWER CAPACITY

The antenna system shall be capable of handling ground transponder peak power outputs of 4000W while operating on a 5 percent duty cycle (equivalent to an average power of 160W).

3.2.8. IMPEDANCE AND VOLTAGE STANDING WAVE RATIO (VSWR)

- a. The design input impedance of the antenna system shall be 50 Ω .
- b. When connected to a 50 Ω source, the TACAN antenna VSWR shall be 1.8:1 or less, across the frequency band.

3.2.9. HARMONIC CONTENT

- a) When measured at an elevation angle of zero to six degrees, the root-sum-square (RSS) value of the second and third harmonics of the 15 Hz component of the radiated signal shall not exceed 25 percent of the 15 Hz component.
- b) The RSS value of the second and third harmonics of the 135 Hz component shall not exceed 25 percent.
- c) The RSS value of the 105, 120, 150, and 165 Hz intermodulation products shall not exceed 25 percent of the 15 Hz component.

3.2.10. ANTENNA MONITORING

3.2.10.1. AZIMUTH ACCURACY MONITORING

- a) Independent of the ground transponder equipment, the antenna system shall detect faults that affect accuracy along any azimuth radial.
- b) Accuracy monitoring shall report an azimuth accuracy fault for any out-of-tolerance condition listed in 3.2.4.
- c) Failure of any part of the monitor itself shall automatically produce the same results as the malfunctioning of the element(s) being monitored.

3.2.10.2. ANTENNA MONITOR PROBES

- a) The antenna shall have internal RF monitoring capability.
- b) The coupling factor shall not vary by more than ± 0.5 dB throughout the range of operational environmental service conditions specified in 3.3.7.2.

3.2.11. HARD ALARM REPORTING

- a) After no more than 10 seconds after the start of an azimuth fault or an out-of-tolerance condition, a TACAN shutdown alarm shall be generated to visual indicators at the antenna control unit and to the TACAN transponder equipment interface.
- b) To avoid unnecessary interruption, the time between detection of an out-of-tolerance azimuth condition and the antenna system monitor taking action shall be user adjustable. *Note that the range of adjustability is a design dependent parameter that will be assessed during the design review process, but in no case shall exceed 10 seconds per 3.2.11-a above.*
- c) The alarm signal shall remain until the alarm condition clears, or a reset is performed.
- d) A reset line shall be provided to the interface by which the antenna system can be reset through the operator's terminal.

3.2.12. REMOTE MAINTENANCE MONITORING

- a) The antenna system shall include an Internet Protocol (IP) enables RMM port.
- b) All available antenna system diagnostic information shall be available at the RMM port.
- c) The antenna reset function shall be available at the RMM port.

3.2.13. EQUIPMENT STARTUP

- a) The time required for the antenna system to fully operate after being turned on from a cold start shall be less than or equal to 45 seconds.
- b) The equipment shall automatically return to normal operation when input power is restored after a power interruption.

3.2.14. POWER CONSUMPTION

The total power consumption by the complete TACAN antenna system shall not exceed 300W.

3.3. PHYSICAL CHARACTERISTICS

The antenna system shall be compatible with AN/URN-32 and AN/URN-33(V) TACAN systems.

3.3.1. RADOME

- a) The antenna shall be enclosed by a radome.
- b) The radome paint shall be Insignia White, FED-STD-595 paint chip 17925.

3.3.2. SIZE

The antenna size, including radome, shall be less than or equal to six (6) feet in diameter and eleven (11) feet in height.

3.3.3. WEIGHT

The total weight of the antenna structure shall be as low as practicable and less than or equal to 900 pounds.

Note: The antenna structure is referencing the unit that is mounted externally to the shelter. The maximum allowed weight does not include interconnecting cables or the antenna control unit mounted inside the equipment shelter.

3.3.4. MOUNTING

- a) The antenna base shall provide for mounting on a steel support ring.
- b) TACAN antenna attachment shall be appropriate to mitigate corrosion issues.
- c) The TACAN antenna shall be provided with attachment points for safely lifting and installing the antenna at its field location.
- d) The antenna shall be permanently marked with a clearly visible north and south orientation mark on an external surface near the bolting circle.
- e) The TACAN antenna mounting interface shall prevent water intrusion.

3.3.5. ANTENNA GROUP ACHIEVED AVAILABILITY (A_A)

Achieved Availability is the average expected availability of the design and includes all preventative and inspection intervals. Typical operational use will be 24 hours per day, 365 days per year for a total of 8760 hours of annual operation. The A_A shall be greater than or equal to 0.99. Achieved availability is measured by the following equation:

$$\frac{\text{Mean Time Between Maintenance (MTBM)}}{\text{MTBM} + \text{Mean Maintenance Time (MMT)}}$$

3.3.5.1. MEAN TIME BETWEEN MAINTENANCE (MTBM)

MTBM is the frequency of Maintenance and is used to determine A_A. It is calculated by dividing the total system operating time by the total number of maintenance events during that time period. Maintenance events that prevent full operation as a result of failures, periodic inspections or preventative maintenance shall be counted toward MTBM to be used in calculating A_A. The MTBM shall be sufficient to meet the availability requirement specified in paragraph 3.3.5.

3.3.5.2. MEAN MAINTENANCE TIME (MMT)

Mean Maintenance Time is a measure of the average hours the system is not operational while undergoing maintenance. For use in calculating A_A, MMT is calculated by dividing the sum of system down time required to perform maintenance by the number of maintenance actions. The Mean Maintenance Time shall be less than or equal to 3 hours.

3.3.6. MAINTAINABILITY

3.3.6.1. PERIODIC INSPECTION AND PREVENTATIVE MAINTENANCE

Periodic inspections and preventative maintenance shall not require the removal of the antenna from the support structure and shall provide maintenance access panels where appropriate.

3.3.6.2. FAULT DETECTION

The antenna group shall have built-in-test capable of detecting greater than 90% of all detectable faults. Rationale for what constitutes an undetectable fault shall be mutually agreed upon by the Government and Contractor.

3.3.6.3. FAULT ISOLATION

The antenna group built-in-test shall have isolation capabilities for all detectable faults. The built-in-test shall isolate 95% of detectable failures to 3 LURs or less. Built-in-test combined with diagnostic tests shall isolate 99% of failures to 1 LRU.

3.3.6.4. FALSE ALARM RATE

A false alarm is an indication of a fault, resulting in a maintenance action, where no fault exists. The Mean Operating Hours Between False Alarms (MOHBFA) for the antenna group shall be greater than 500 hours. MOHBFA is calculated by dividing the total operating hours by the number of false alarms.

3.3.7. MODULAR CONSTRUCTION

- a) TACAN antenna system repairs shall be accomplished onsite using Line Replaceable Assets (LRAs).
- b) LRA replacement shall not require unmounting the antenna from the support structure.

3.3.8. SOFTWARE/FIRMWARE UPDATES

- a) Software and firmware for the TACAN antenna system shall be field updatable, without adding or modifying existing hardware components.
- b) The software and firmware shall use less than 50% of the available volatile and non-volatile memory.

3.3.9. ENVIRONMENTAL CONDITIONS

The TACAN antenna system operating and non-operating environmental conditions were derived from MIL-STD-810H.

3.3.9.1. NON-OPERATING CONDITIONS

3.3.9.1.1. NON-OPERATING TEMPERATURE RANGE

The antenna system packaged for storage, shipping or transporting (non-operating) shall withstand a temperature range of -46°C to 71°C.

3.3.9.1.2. NON-OPERATING ALTITUDE

The antenna system packaged for storage, shipping or transporting (non-operating) shall withstand atmospheric pressure of 575 to 1025 millibars (mb) (equivalent to 15,000 feet above sea level in standard atmosphere down to 300 feet below sea level).

3.3.9.1.3. VIBRATION

The antenna system shall not be damaged when subjected to the vibrations listed in Table 514.8C-I, Method 514.8 Annex C of MIL-STD-810H and the exposure duration provided in Annex C paragraph 2.1.4.

3.3.9.1.4. SHOCK

The antenna system shall be capable of withstanding the shock test specified in Method 516.8, Procedure IV, Logistic Drop Test, of MIL-STD-810H.

3.3.9.2. OPERATING CONDITIONS

3.3.9.2.1. OUTDOOR TEMPERATURE

All outdoor equipment shall operate in the temperature range of -46°C to 49°C.

3.3.9.2.2. INDOOR TEMPERATURE

All indoor equipment shall operate in the temperature range of -18°C to 38°C.

3.3.9.2.3. THERMAL DESIGN

Only convective or natural cooling methods shall be used.

3.3.9.2.4. RAIN

All outdoor equipment shall operate while exposed to wind-blown rain, at a rate of 1.7mm/minute (4 inches/hour), and up to 18 m/s (40 mph) blowing wind.

3.3.9.2.5. OPERATING ALTITUDE

All indoor and outdoor equipment shall operate over the atmospheric pressure range of 700 to 1025 mb (equivalent to 10,000 feet above sea level in standard atmosphere down to 300 feet below sea level).

3.3.9.2.6. WIND VELOCITY

The antenna shall be capable of withstanding wind conditions up to 150 mph velocity.

3.3.9.2.7. ICE LOADING

The antenna shall withstand up to ½ inch accumulation of clear ice on all outdoor components.

3.3.9.2.8. HUMIDITY

- a) Equipment shall operate within a relative humidity range from 5 to 100 percent when the temperature is 40°C or less.
- b) Above 40°C, the equipment shall operate with a relative humidity based upon a dew point of 40°C.

3.3.9.2.9. FINE SAND (DUST)

The TACAN antenna system outdoor equipment shall be impervious to sand and dust intrusion.

3.3.9.3. ELECTROMAGNETIC ENVIRONMENTAL EFFECTS (E3)

3.3.9.3.1. MARGINS

Margins shall be established for safety and mission critical subsystems/equipment within the antenna system. Margins shall be no less than 6 dB for safety critical subsystems/equipment in accordance with MIL-STD-464D, paragraph 5.1.

3.3.9.3.2. INTRA-SYSTEM ELECTROMAGNETIC COMPATIBILITY (EMC)

The antenna system shall be electromagnetically compatible within itself such that the system operational performance requirements are met in accordance with MIL-STD-464, paragraph 5.2.

3.3.9.3.3. EXTERNAL RF EME

The antenna system shall be electronically compatible with its defined external RF EME such that its system operational requirements are met in accordance with MIL-STD-464D, paragraph 5.3, Table IV.

3.3.9.3.4. LIGHTNING

The system shall meet its operational performance requirements when subjected to direct, indirect, and near strike lightning effects in accordance with MIL-STD-464D, paragraph 5.5.

3.3.9.3.5. ELECTROMAGNETIC INTERFERENCE (EMI)

The antenna system shall meet the applicable EMI requirements within MIL-STD-461G for Navy ground systems to include CE102, CS101, CS114, CS115, CS116, CS117, CS118, RE102, and RS103.

3.3.9.3.6. ELECTROSTATIC ENVIRONMENTS

The antenna system shall safely control and dissipate the build-up of electrostatic charges in accordance with MIL-STD-464D, paragraph 5.8.

3.3.9.3.7. RADIATION HAZARDS

The antenna system design shall protect personnel, fuels, and ordnance from hazardous effects of electromagnetic radiation in accordance with MIL-STD-464D, paragraph 5.9.

3.3.9.3.8. LIFE CYCLE E3 HARDNESS

The system operational performance and E3 requirements shall be met throughout the rated life cycle of the system in accordance with MIL-STD-464D, paragraph 5.10.

3.3.9.3.9. ELECTRICAL BONDING

The antenna system shall include the necessary electrical bonding to meet the E3 requirements in accordance with MIL-STD-464D, paragraph 5.11.

3.3.9.3.10. ELECTROMAGNETIC SPECTRUM SUPPORTABILITY

The antenna system shall comply with the DoD, National, and International Spectrum regulations for the use of the electromagnetic spectrum in accordance with MIL-STD-464D, paragraph 5.15.

3.3.10. INTERCONNECTING CABLING

- a) Cables and connectors, excluding the ground cable, shall be protected from exposure to the elements.
- b) All connections to the antenna shall be via a connector or an internal terminal strip at the base along the centerline of the antenna.
- c) All connectors and terminal strips shall be recessed or otherwise located such that no damage shall occur when the antenna is placed on a flat surface.
- d) A grounding lug to accommodate 6 American Wire Gauge (AWG) copper wire shall be provided near the bottom center of the antenna mounting base.
- e) All interconnecting cables between the antenna and the antenna control unit shall terminate at one or more connectors. If more than one connector is used, they shall not be interchangeable.

3.3.11. INPUT POWER

3.3.11.1. AC POWER

The antenna system shall operate from 120 VAC, 50-400 Hz, single phase power as the primary power source.

3.3.11.2. LEAKAGE CURRENTS

Leakage current shall be less than 5 mA.

3.3.11.3. POWER FACTOR

The power factor for the TACAN antenna system shall be within 0.7(lag) – 0.7(lead).

3.3.11.4. GROUNDING

The ground reference, shielding and bonding for the antenna system shall be in accordance with MIL-HDBK-1857.

3.3.12. MARKING

The antenna and control indicator shall have nameplates and marking in accordance with MIL-STD-130N.

3.3.13. SAFETY

3.3.13.1. PERSONNEL SAFETY

- a) Personnel shall be protected from injury by sharp edges of equipment or cabinets during maintenance.
- b) All equipment enclosures shall be designed and install so that the enclosure and the mounted LRU and/or components will remain upright and in place and all access panels and drawers shall remain secure to the enclosure.
- c) The materials install in the equipment shall not liberate gases that when combined with the atmosphere form an acid or corrosive alkali, nor shall they generate toxic or corrosive dusts, mists, or fumes which would be detrimental to the performance of the equipment or health of the equipment operators.
- d) The materials shall not liberate gases that will produce a flammable or explosive atmosphere.
- e) Mercury or asbestos shall not be used.

3.3.13.2. ELECTRICAL SAFETY

- a) A means shall be provided to protect personnel from accidental contact with voltages in excess of 30 V_{rms} or DC during normal operations and maintenance of the equipment.
- b) After power to the equipment is turned off, the system shall discharge all capacitors to 30 V or less within 2 seconds.
- c) Personnel shall be protected from circuits carrying currents greater than 21 mA at 10.5 VAC, and 80 mA at 40 VDC.
- d) The hot side of the incoming power line switch connections shall be given physical protection against accidental contact.

3.3.14. CONTROL INDICATOR DISPLAY

The control indicator display shall meet the requirements in paragraph 5.2.3 of MIL-STD-1472G.

Attachment 1

Transponder Group, TACAN AN/URN-32 Cable Running Sheets